

Milestone 2 Submission

Rethinking Chain-of-Thought Reasoning through Diffusion Language Models: A Comparative Study with Auto-Regressive Architectures

Outline, Figures, and Tables Draft updated with complete verified results

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1. Thesis outline

1. Introduction: motivate the registered thesis question and explain why diffusion generation is a meaningful alternative to autoregressive decoding.
2. Related work: discrete diffusion language modeling, open autoregressive baselines, reasoning benchmarks, planning benchmarks, and pass@k evaluation.
3. Methodology: benchmarks, checkpoints, prompt modes, inference backends, answer extraction, deterministic versus stochastic evaluation, and the per-question correlation protocol.
4. Results: deterministic planning, pass@k ranking reversal, GSM8K prompt/few-shot sensitivity, Trip Planning generalization, hyperparameter analysis, and per-question correlation analysis.
5. Discussion: why diffusion models have high pass@1 but slower pass@k growth, why autoregressive models scale more strongly with sampling, and why the two paradigms appear complementary.
6. Conclusion and future work: the answer to “which paradigm is better?”

2. Introduction

Large language models are typically discussed through the lens of autoregressive decoding, where tokens are generated left-to-right one step at a time. Diffusion language models instead refine a partially masked sequence through iterative denoising, making them a fundamentally different candidate for reasoning and planning [1-5].

This milestone asks the thesis question stated in the registration document directly: can chain-of-thought reasoning be rethought through diffusion language models, and if so under what evaluation protocol? The empirical comparison covers two diffusion families (LLaDA-8B and Dream-v0-7B) and two autoregressive families (Qwen2.5-7B and Llama-3.1-8B) across GSM8K, Countdown-cd4, and Trip Planning under both deterministic and stochastic pass@k evaluation [2-8].

The current milestone have 3 contributions. First, diffusion checkpoints are strongest on deterministic or pass@1 planning settings, but autoregressive models overtake at larger k. Second, per-question Pearson correlations show that failure-mode alignment is more strongly organized by generation paradigm than by shared pretrained architecture. Third, prompt sensitivity differs by paradigm: diffusion models are much stronger at zero-shot GSM8K, whereas autoregressive models benefit more from few-shot examples.

3. Related work

The diffusion side of the thesis builds on structured denoising diffusion models in discrete state spaces and on recent large language diffusion systems such as LLaDA and Dream 7B [1-3]. These works make token-level denoising a viable large-scale alternative to autoregressive decoding and motivate the central comparison in this thesis.

The autoregressive baselines are intentionally strong open checkpoints at a similar parameter scale. Qwen2.5 and Llama 3.1 provide both base and instruction-tuned variants and serve as practical reference systems for reasoning and planning tasks [4,5]. On the evaluation side, GSM8K remains a standard benchmark for multi-step mathematical reasoning, TravelPlanner provides a realistic

planning benchmark, and the pass@k estimator follows the evaluation protocol popularized in HumanEval [6-8].

Because inference speed is part of the practical trade-off, the milestone also distinguishes fast-dLLM from slower dLLM baselines on the diffusion side and vLLM with PagedAttention on the autoregressive side [9]. This is important for interpreting both deterministic parity checks and the feasibility of large pass@k sweeps.

4. Methodology structure

The methodology chapter is organized around a single evaluation pipeline so that model-family differences can be interpreted under controlled prompts, answer extraction, and reporting. The same benchmarks, prompt constructions, and aggregation rules are reused across all four model families to keep the comparison focused on the generation paradigm rather than on incompatible tooling choices.

Table 1. Benchmark suite and experimental role in the thesis narrative.

Component	Role in the thesis	Evaluation mode	Notes
GSM8K	Reasoning sanity check and prompt/few-shot sensitivity benchmark	Deterministic and pass@k; base and instruct settings	Near-saturated at 7-8B scale, but informative for prompt sensitivity
Countdown-cd4	Core arithmetic planning benchmark and central ranking-reversal evidence	Deterministic validation plus pass@k (n=128)	Main benchmark for the pass@k crossover narrative
Trip Planning	Structured natural-language planning benchmark	Deterministic comparison plus pass@k (n=64)	Tests whether Countdown patterns generalize beyond arithmetic planning
Per-question correlation analysis	Novel analysis of failure-mode alignment	Pearson r on per-question accuracy vectors	128 samples/model on Countdown and GSM8K; 64 on Trip Planning

Table 2. Model families and inference backends used in Milestone 2.

Family	Base	Instruct	Params	Type
LLaDA	GSAI-ML/LLaDA-8B-Base	GSAI-ML/LLaDA-8B-Instruct	8B	Diffusion (masked)
Dream	Dream-org/Dream-v0-Base-7B	Dream-org/Dream-v0-Instruct-7B	7B	Diffusion (masked, Qwen2.5 backbone)
Qwen	Qwen/Qwen2.5-7B	Qwen/Qwen2.5-7B-Instruct	7B	Autoregressive
Llama	meta-llama/Llama-3.1-8B	meta-llama/Llama-3.1-8B-Instruct	8B	Autoregressive

Shared evaluation pipeline used for all diffusion and autoregressive checkpoints

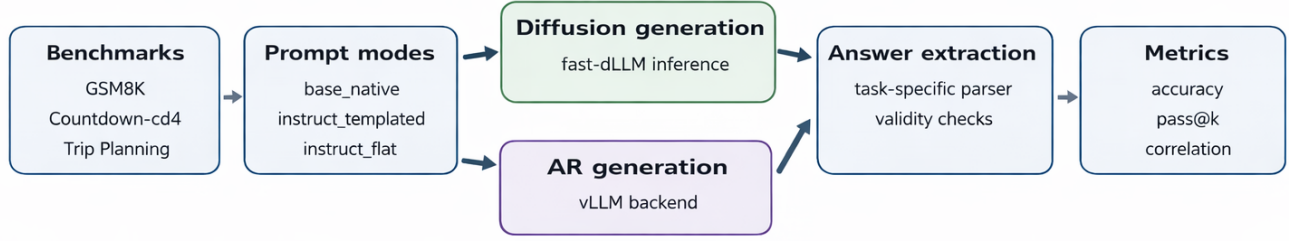


Figure 1. Shared evaluation pipeline for diffusion and autoregressive experiments. The same benchmark definitions, prompt modes, generation stage, answer extraction, and metrics are used to support comparable analysis across all four model families.

- Prompt modes: `base_native`, `instruct_templated`, and `instruct_flat`. The templated instruct setting is the default instruct evaluation; `instruct_flat` is an explicit prompt-format ablation.
- Evaluation modes: deterministic accuracy uses `temperature=0` and `n_samples=1`; `pass@k` uses `temperature=0.7` with `n=16, 64, or 128` depending on benchmark and setup.
- Inference provenance: all main dLLM comparisons use fast-dLLM, all main AR comparisons use vLLM, and standard dLLM / AR-HF runs are retained only for targeted deterministic validation.
- Correlation protocol: per-question Pearson correlations are computed from question-level accuracy vectors using 128 samples/model on Countdown and GSM8K and 64 samples/model on Trip Planning.

5. Results section structure: topic sentences and evidence

As required by the milestone instructions, the results chapter is first outlined through topic sentences and the evidence needed to support each one. Tables 3a and 3b are the central milestone artifacts: a reader should be able to reconstruct the thesis story from these topic sentences alone.

Table 3a. Results subsections and supporting evidence (part 1).

Proposed subsection	Topic sentence	Required evidence	Status
5.1 Deterministic planning	Diffusion models win deterministic planning evaluations on Countdown and Trip Planning.	Countdown deterministic comparison; Trip Planning deterministic table; short runtime interpretation.	Included
5.2 Countdown pass@k crossover	On Countdown base models, diffusion checkpoints lead at pass@1 but autoregressive models overtake at larger k.	Countdown base crossover plot; pass@1 summary table; narrative highlighting pass@1 and pass@128 ordering.	Included
5.3 Countdown instruct sensitivity	Countdown instruct results are highly model-dependent and show that instruct tuning helps AR models more consistently than dLLMs.	Countdown instruct plot; prompt-mode diagnostic plot; pass@1 summary table.	Included
5.4 GSM8K prompt and few-shot sensitivity	GSM8K is near-saturated, but zero-shot versus few-shot behavior differs strongly by paradigm.	GSM8K pass@k panels; prompt-format figure; pass@1 summary table.	Included

Table 3b. Results subsections and supporting evidence (part 2).

Proposed subsection	Topic sentence	Required evidence	Status
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Proposed subsection	Topic sentence	Required evidence	Status
5.5 Trip Planning generalization	Trip Planning reproduces the ranking-reversal story while showing especially strong complementarity between paradigms.	Trip Planning pass@k plot; deterministic comparison; correlation matrices.	Included
5.6 Hyperparameter analysis	Hyperparameter tuning improves absolute quality but does not remove the need for a family-level pass@k comparison.	Best pass@1 table; best pass@128 plot; key-finding bullets.	Included
5.7 Per-question correlation analysis	Failure-mode alignment depends more on generation paradigm than on shared pretrained architecture in key settings.	Six correlation matrices; cross-benchmark summary table; correlation summary figure; HTML placeholder figure.	Included
5.8 Discussion bridge	The two paradigms are complementary: dLLMs are stronger in single-shot settings, while AR models benefit more from large-k search.	Cross-benchmark synthesis paragraph; milestone take-home messages.	Included

6. Draft figures and tables plan

Table 4a. Planned figures for Milestone 2.

Artifact	Draft caption / thesis role	Status
Figure 1. Shared evaluation pipeline	Unified schematic of benchmarks, prompting, generation, answer extraction, and metrics.	Included
Figure 2. GSM8K pass@k panels	Shows the difference between base, zero-shot instruct, and four-shot instruct scaling.	Included
Figure 3. GSM8K prompt-format sensitivity	Shows templated versus flat prompting for instruct checkpoints and highlights the Llama-Instruct drop from 78.22% to 21.05% at pass@1.	Included
Figure 4. Countdown base pass@k crossover (R1)	Central thesis figure: diffusion models start higher but AR models scale more steeply and overtake around k=8-16.	Included
Figure 5. Countdown instruct pass@k	Supports the claim that Dream-Instruct remains competitive while LLaDA-Instruct collapses.	Included
Figure 6. Countdown prompt-mode diagnostic	Prompt-format ablation from earlier smaller instruct runs.	Included

Table 4b. Planned tables and remaining evidence artifacts for Milestone 2.

Artifact	Draft caption / thesis role	Status
Figure 7. Trip Planning pass@k	All-model pass@k view for the structured planning benchmark.	Included
Figure 8. Hyperparameter sweep summary	Best pass@128 by model/configuration from the GSM8K sweep.	Included
Figure 9. Correlation summary	Cross-benchmark bar chart comparing within-paradigm and strongest cross-paradigm correlations.	Included
Figure 10. HTML comparison placeholder (R2)	Placeholder layout for the six interactive per-question comparison views named in the draft.	Included
Tables 5-17	Deterministic/pass@1 summaries, hyperparameter table, and the six correlation matrices plus summary table.	Included

7. Current empirical evidence

The figures and tables below provide the milestone-ready empirical evidence supporting the topic sentences above. Pass@k result grids are plotted directly from the exact draft values, while compact deterministic/pass@1 tables preserve the single-shot results in a format that is easy to scan.

7.1 GSM8K: reasoning sanity check and prompt/few-shot sensitivity

GSM8K is the least decisive benchmark for the final ranking because all models approach saturation at larger k. Its most useful role in this milestone is therefore diagnostic: it shows that diffusion models are much stronger at zero-shot instruct prompting, whereas autoregressive models benefit more from few-shot examples and can catch up or overtake under stronger prompting.

Table 5. GSM8K deterministic/pass@1 summary across base and instruct configurations.

Condition	LLaDA	Dream	Qwen	Llama
Base, 0-shot (n=128)	63.14%	72.39%	70.93%	14.33%
Base, 4-shot (n=16)	66.48%	71.34%	72.27%	14.53%
Instruct, 0-shot (n=16)	74.85%	73.39%	53.37%	48.78%
Instruct, 4-shot (n=16)	77.47%	76.15%	78.88%	78.20%
Instruct templated, 4-shot (n=128)	75.57%	76.15%	79.77%	78.22%
Instruct flat, 4-shot (n=128)	76.90%	70.93%	73.83%	21.05%

GSM8K pass@k curves across base and instruct settings

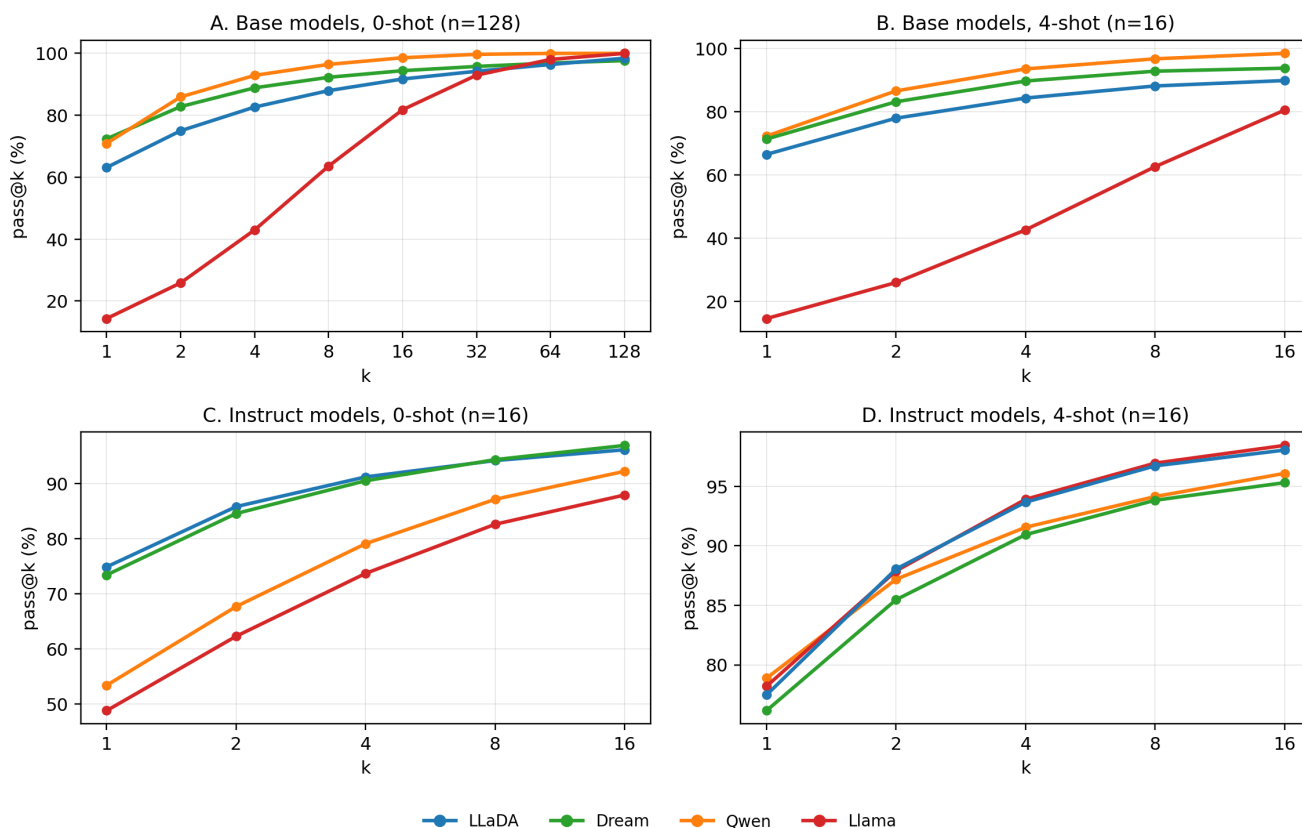


Figure 2. GSM8K $\text{pass}@k$ curves across base and instruct settings. Panels A-D show that GSM8K is near-saturated at 7-8B scale, but they also reveal a clear paradigm difference in prompt sensitivity: diffusion instruct models start much higher at 0-shot, whereas AR instruct models gain more from 4-shot prompting. Source: redrawn from MILESTONE_2_DRAFT.md Sections 3.1, 3.2, 3.4, and 3.5.

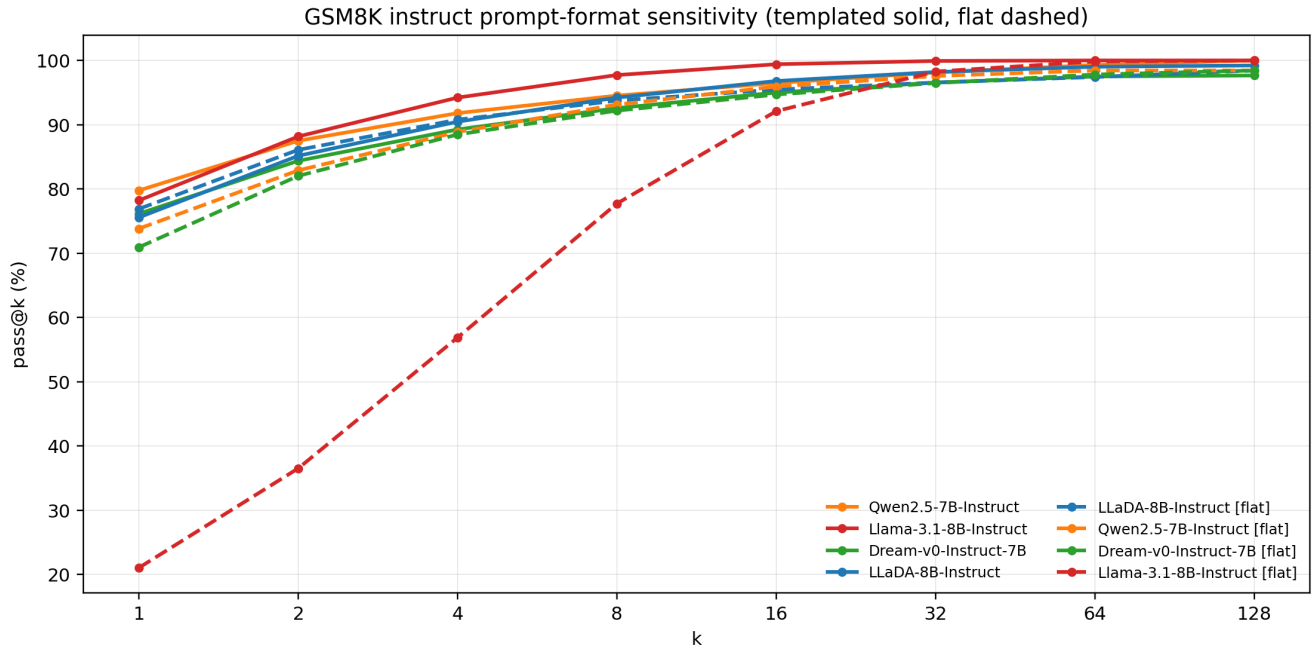


Figure 3. GSM8K instruct prompt-format sensitivity at $n=128$. Templated prompts (solid) and flat prompts (dashed) behave similarly for most checkpoints, but Llama-3.1-8B-Instruct drops from 78.22% to 21.05% at $\text{pass}@1$ when the prompt is flattened.

The exact $\text{pass}@1$ values make the paradigm difference especially clear. At 0-shot instruct, LLaDA-8B-Instruct reaches 74.85% and Dream-v0-Instruct-7B reaches 73.39%, versus 53.37% for Qwen2.5-7B-Instruct and 48.78% for Llama-3.1-8B-Instruct. At 4-shot instruct, the gap closes: Qwen rises to 78.88% and Llama to 78.20%, while LLaDA reaches 77.47% and Dream 76.15%. This supports the thesis claim that AR models depend more strongly on in-context examples, whereas dLLMs have stronger zero-shot generalization on GSM8K.

7.2 Countdown-cd4: core planning benchmark

Countdown is the central Milestone 2 benchmark because it makes the evaluation-protocol story unambiguous. Deterministically, diffusion models are much stronger; under $\text{pass}@k$, autoregressive models scale much more steeply and reverse the ranking at larger k .

Table 6. Countdown deterministic comparison.

Model	Method	Acc (bs=1)	Time (bs=1)	Acc (bs=8)	Time (bs=8)	Speedup
LLaDA-Base	Fast-dLLM	13.7%	16m 9s	14.5%	5m 19s	3.03×
LLaDA-Base	dLLM	15.4%	25m 31s	14.7%	30m 1s	0.85×
Dream-Base	Fast-dLLM	14.4%	15m 46s	14.1%	6m 46s	2.33×
Dream-Base	dLLM	14.6%	23m 39s	14.3%	18m 35s	1.27×

Model	Method	Acc (bs=1)	Time (bs=1)	Acc (bs=8)	Time (bs=8)	Speedup
Qwen-Base	vLLM	6.0%	6m 35s	6.0%	51.6s	7.65×
Qwen-Base	AR	6.0%	13m 46s	6.0%	2m 36s	5.31×
LLaDA-Inst	Fast-dLLM	1.5%	16m 0s	1.9%	5m 38s	2.84×
LLaDA-Inst	dLLM	1.9%	25m 56s	1.8%	30m 50s	0.84×
Dream-Inst	Fast-dLLM	7.5%	12m 16s	7.9%	6m 4s	2.02×
Dream-Inst	dLLM	3.5%	23m 43s	3.4%	19m 0s	1.25×
Qwen-Inst	vLLM	2.6%	6m 33s	2.8%	51.9s	7.58×
Qwen-Inst	AR	1.0%	14m 2s	0.9%	2m 43s	5.16×

Table 7. Countdown pass@1 summary across base, instruct-templated, and instruct-flat settings.

Setting	LLaDA	Dream	Qwen	Llama
Base pass@1 (n=128)	14.29%	11.30%	4.04%	2.70%
Instruct templated pass@1 (n=128)	2.41%	9.15%	7.84%	5.78%
Instruct flat pass@1 (earlier smaller run)	2.84%	1.81%	1.05%	4.99%

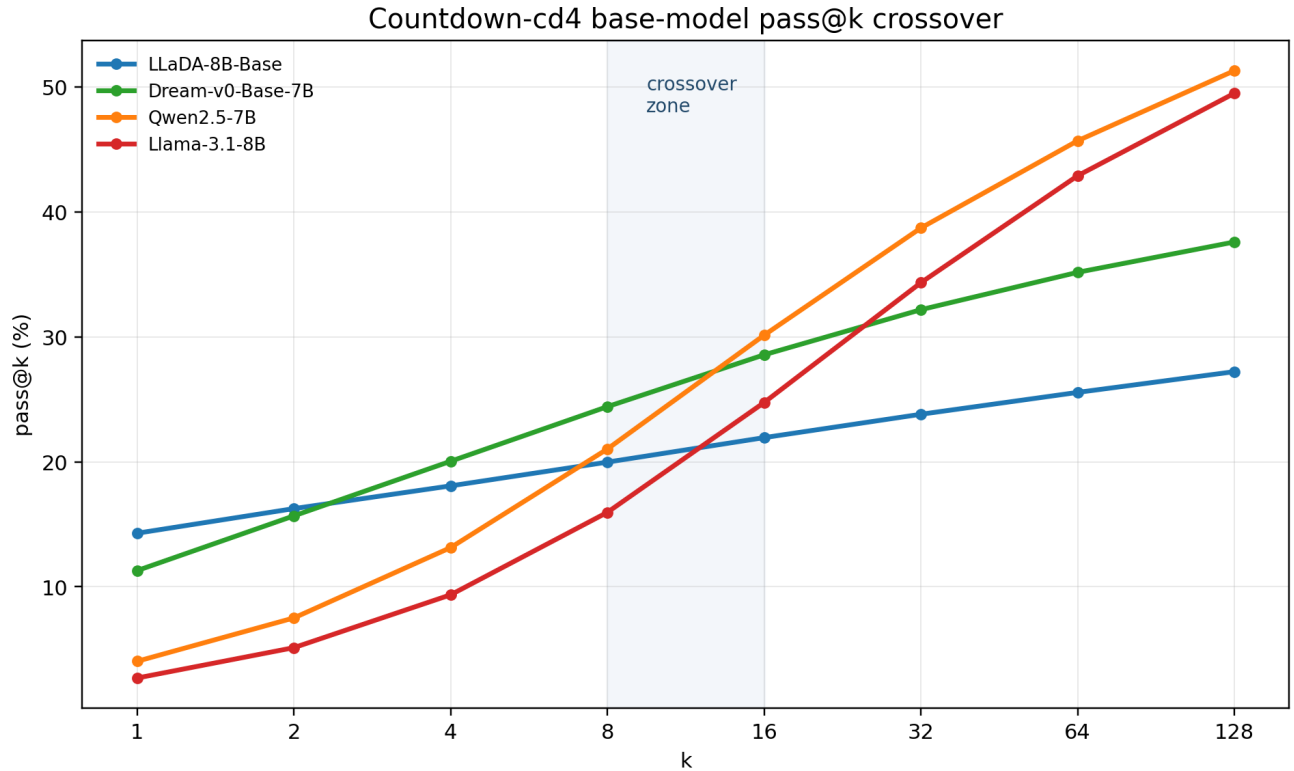


Figure 4. Countdown-cd4 base-model pass@k crossover (R1). Diffusion models start higher at low k , but autoregressive models grow more steeply and overtake around $k=8-16$, making this the central ranking-reversal figure of the thesis.

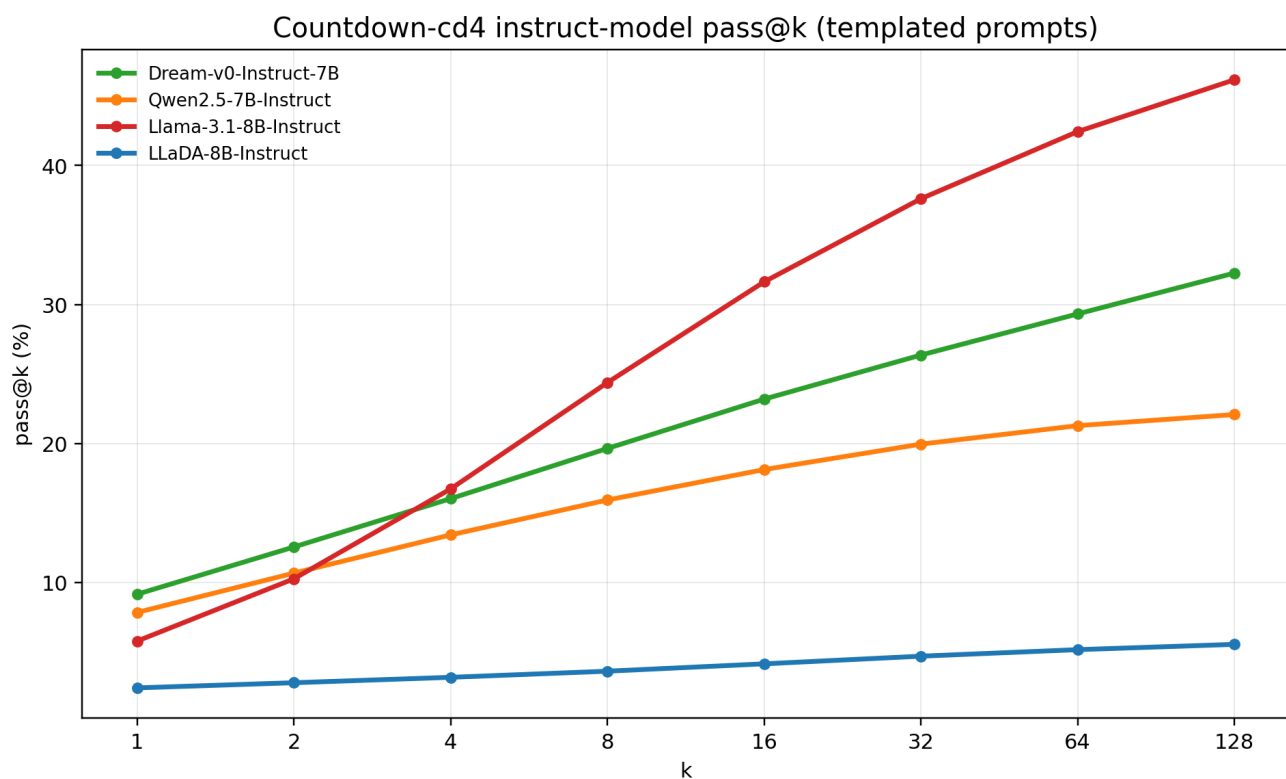


Figure 5. Countdown-cd4 instruct-model pass@k under templated prompting. Dream-v0-Instruct-7B remains competitive, Llama-3.1-8B-Instruct becomes the strongest high-k model, and LLaDA-8B-Instruct remains near-collapsed throughout the curve.

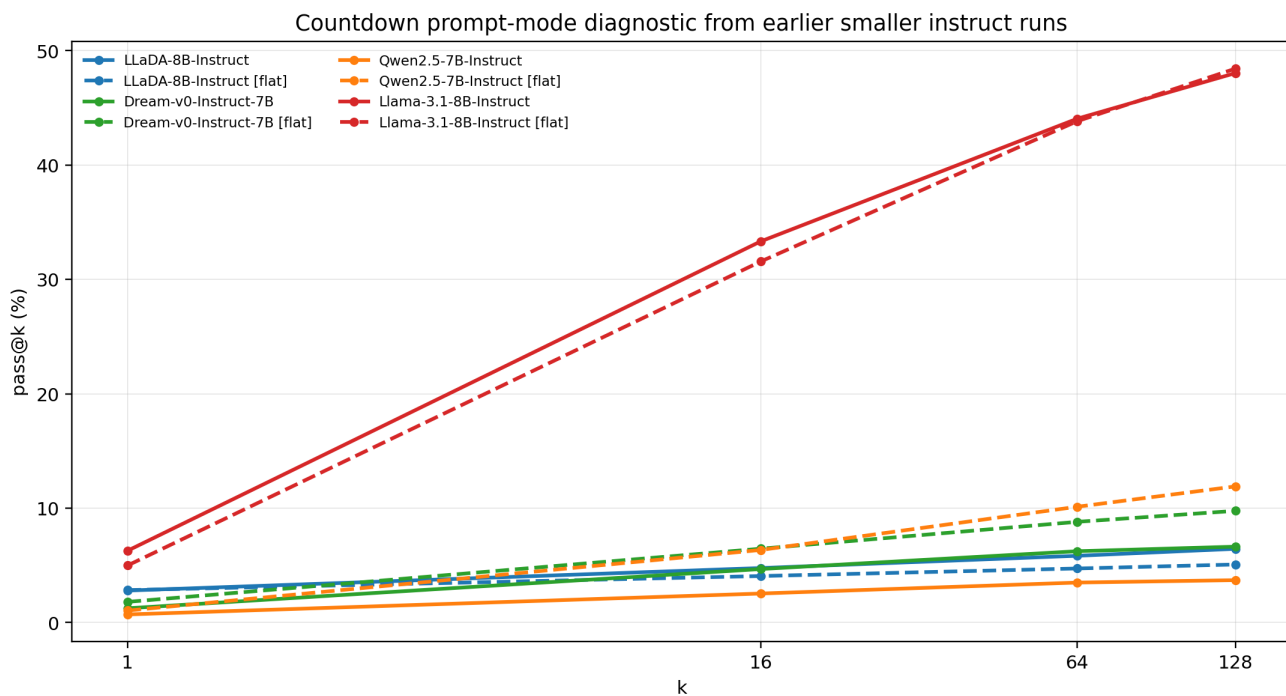


Figure 6. Countdown prompt-mode diagnostic from earlier smaller instruct runs. Prompt formatting changes Dream and Qwen substantially, but it barely changes the already-strong Llama-Instruct curve and does not rescue LLaDA-Instruct.

The exact base-model ordering is the key result. At pass@1, LLaDA (14.29%) > Dream (11.30%) > Qwen (4.04%) > Llama (2.70%), so diffusion models dominate. At pass@128, the ranking reverses to Qwen (51.31%) > Llama (49.50%) > Dream (37.60%) > LLaDA (27.22%). The crossover occurs around pass@8-16 for Dream versus the AR baselines and around pass@16-32 for LLaDA versus the AR baselines. This is the clearest evidence that dLLMs concentrate probability mass into a smaller set of strong samples, whereas AR models maintain broader diversity and therefore scale more strongly under pass@k evaluation.

The deterministic table adds the practical runtime story. Fast-dLLM preserves much of the diffusion quality while materially reducing latency relative to standard dLLM, but vLLM remains much faster for the autoregressive checkpoints. In the instruct setting, Dream-Instruct is the only dLLM checkpoint that remains competitive; LLaDA-Instruct reaches only 5.54% at pass@128 under templated prompting.

7.3 Trip Planning: structured natural-language planning

Trip Planning generalizes the Countdown narrative to a more natural planning benchmark. Dream remains strongest at low-k and deterministic settings, but the flatter autoregressive distributions scale more aggressively under sampling and become strongest at larger k, especially for Llama flat prompting.

Table 8. Trip Planning deterministic comparison.

Model / Variant	Accuracy
Dream-v0-Instruct-7B [bl=256] [dllm]	15.00%
Dream-v0-Base-7B [bl=256] [dllm]	14.50%
Dream-v0-Base-7B [bl=32] [fast-dllm]	13.50%
Dream-v0-Instruct-7B [bl=32] [fast-dllm]	12.50%
Dream-v0-Instruct-7B [flat] [bl=32] [fast-dllm]	12.50%
Dream-v0-Base-7B [bl=256] [dllm]	14.50%
LLaDA-8B-Base [bl=256] [dllm]	11.50%
Llama-3.1-8B-Instruct [flat] [vllm]	9.50%
LLaDA-8B-Instruct [bl=256] [dllm]	8.00%
LLaDA-8B-Base [bl=32] [fast-dllm]	5.00%
Llama-3.1-8B [vllm]	4.50%
LLaDA-8B-Instruct [bl=32] [dllm]	4.50%
Qwen2.5-7B [vllm]	1.50%
Qwen2.5-7B-Instruct [vllm]	1.50%
Llama-3.1-8B-Instruct [vllm]	2.00%
Qwen2.5-7B-Instruct [flat] [vllm]	2.00%

Table 9. Trip Planning pass@1 summary for all stochastic milestone runs.

Model	pass@1
Dream-v0-Base-7B [fast-dllm]	5.44%
Dream-v0-Instruct-7B [fast-dllm]	5.66%
LLaDA-8B-Base [fast-dllm]	3.55%
LLaDA-8B-Instruct [fast-dllm]	1.04%
LLaDA-8B-Instruct [flat] [fast-dllm]	1.24%
Llama-3.1-8B [vllm]	2.50%
Llama-3.1-8B-Instruct [vllm]	1.66%
Llama-3.1-8B-Instruct [flat] [vllm]	6.00%
Qwen2.5-7B [vllm]	1.48%
Qwen2.5-7B-Instruct [vllm]	1.65%

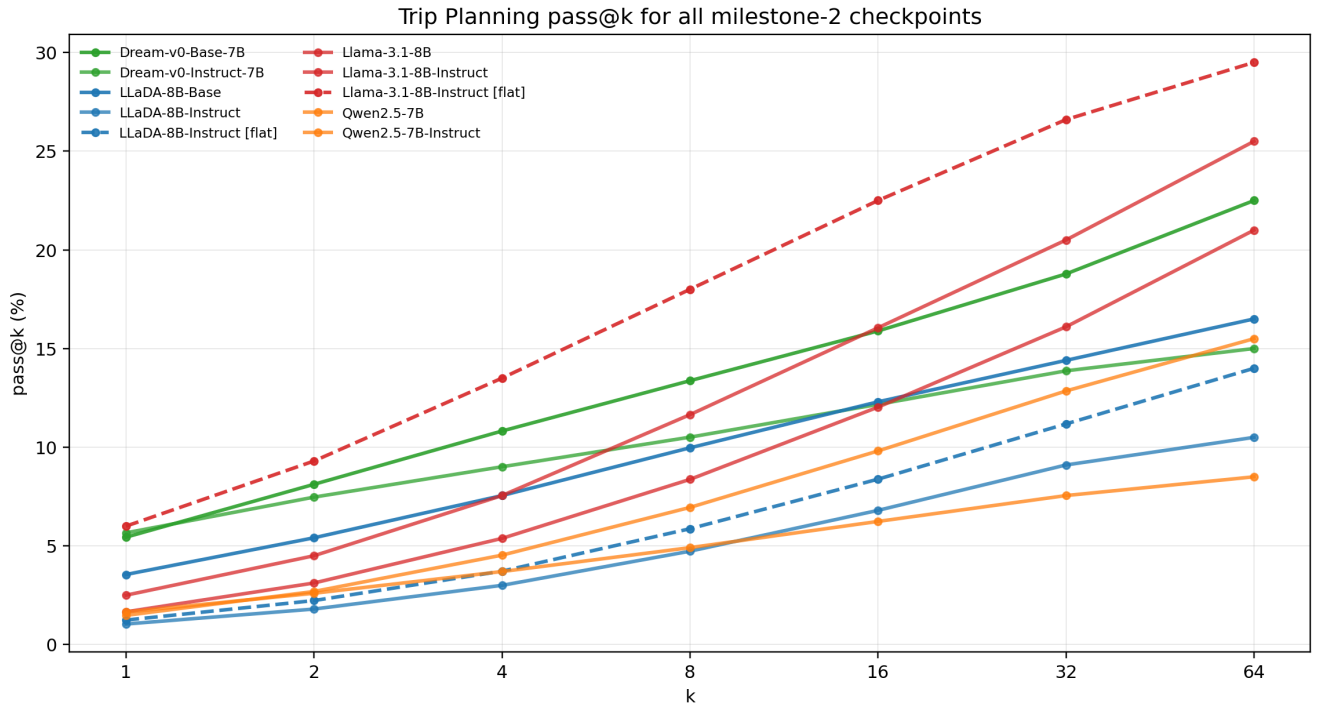


Figure 7. Trip Planning pass@k for all milestone-2 checkpoints. Dream-v0-Base-7B leads the standard checkpoints at pass@1, while Llama-3.1-8B-Instruct with flat prompting becomes the strongest high-k curve and reaches 29.50% at pass@64.

The pass@k table is now visualized as a plot to emphasize the growth-rate comparison. Dream-v0-Base-7B starts at 5.44% pass@1 and reaches 22.50% at pass@64, whereas Llama-3.1-8B-Instruct [flat] starts at 6.00% and reaches 29.50% by pass@64. Standard Qwen and Llama base checkpoints start low but scale steadily, while LLaDA-Instruct remains weak even after switching to flat prompts. Together with Countdown, this supports the thesis claim that diffusion models offer stronger single-shot planning quality, while autoregressive models offer stronger search behavior when multiple samples are allowed.

7.4 Hyperparameter analysis and the pivot away from tuning

Before settling on the comparative pass@k story, the project conducted an extensive GSM8K hyperparameter sweep for LLaDA-8B, Dream-7B, and Qwen2.5-7B. The sweep matters because it justifies why the thesis pivots away from “find the single best dLLM configuration” and toward “compare paradigms under a controlled evaluation protocol.”

Table 10. Best pass@1 configurations from the GSM8K hyperparameter sweep.

Model	Best pass@1 Config	pass@1	Time
LLaDA-8B	g256_s256_b128	71.01%	14.5 min
Dream-7B	g256_s32_b32	70.66%	11.8 min
Qwen2.5-7B	g256 (vLLM)	72.97%	5.7 min

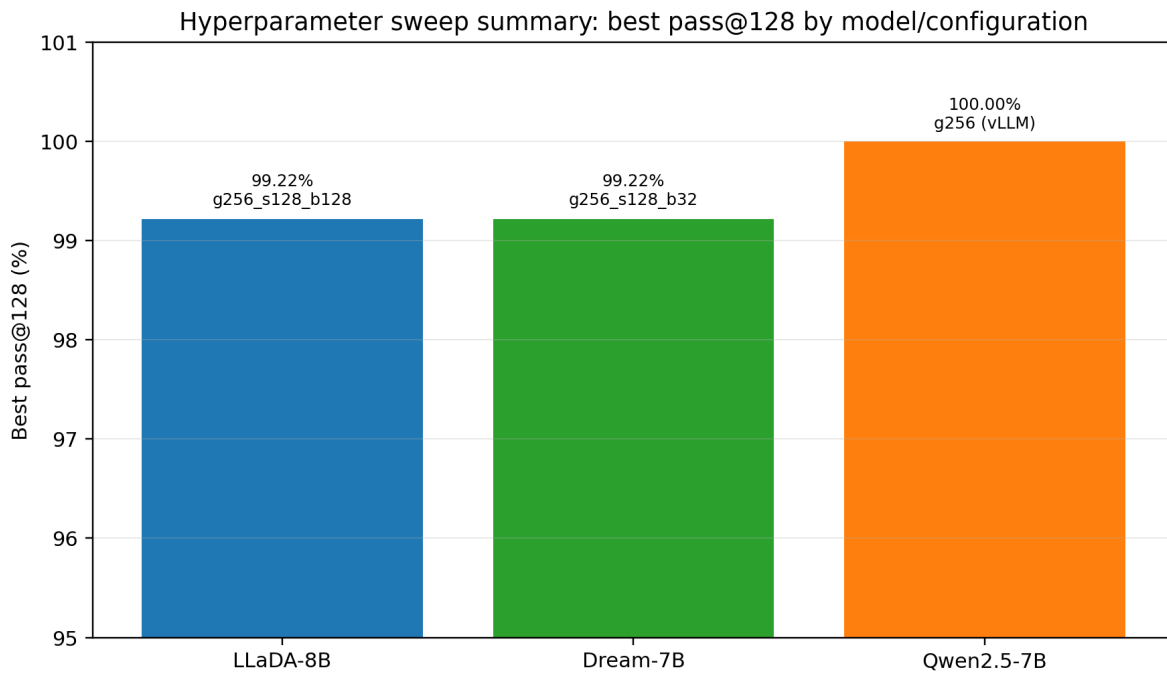


Figure 8. Hyperparameter sweep summary for best pass@128 by model and configuration. Even after tuning, the sweep does not produce a single dominant diffusion configuration that would remove the need for a paradigm-level comparison.

- Generation length is the most impactful parameter. Moving from 128 to 256 tokens improves pass@1 by roughly 20-25 percentage points.
- Steps show diminishing returns: 128-256 steps are the useful range, while 32 steps significantly underperform.
- Block length has nuanced effects; smaller block lengths such as 32 often improve the quality-vs-latency trade-off.
- No single configuration dominates across pass@1, pass@128, and runtime. This motivates the thesis pivot from hyperparameter hunting to controlled comparative pass@k analysis.

7.5 Per-question correlation analysis (novel milestone contribution)

The most novel addition in this revised milestone draft is the per-question correlation analysis. For each question, accuracy is computed as the fraction of correct samples for that model, and Pearson correlation is then measured between model pairs across questions. This turns the comparison from “which model scores higher on average?” into “which models succeed on the same questions?”

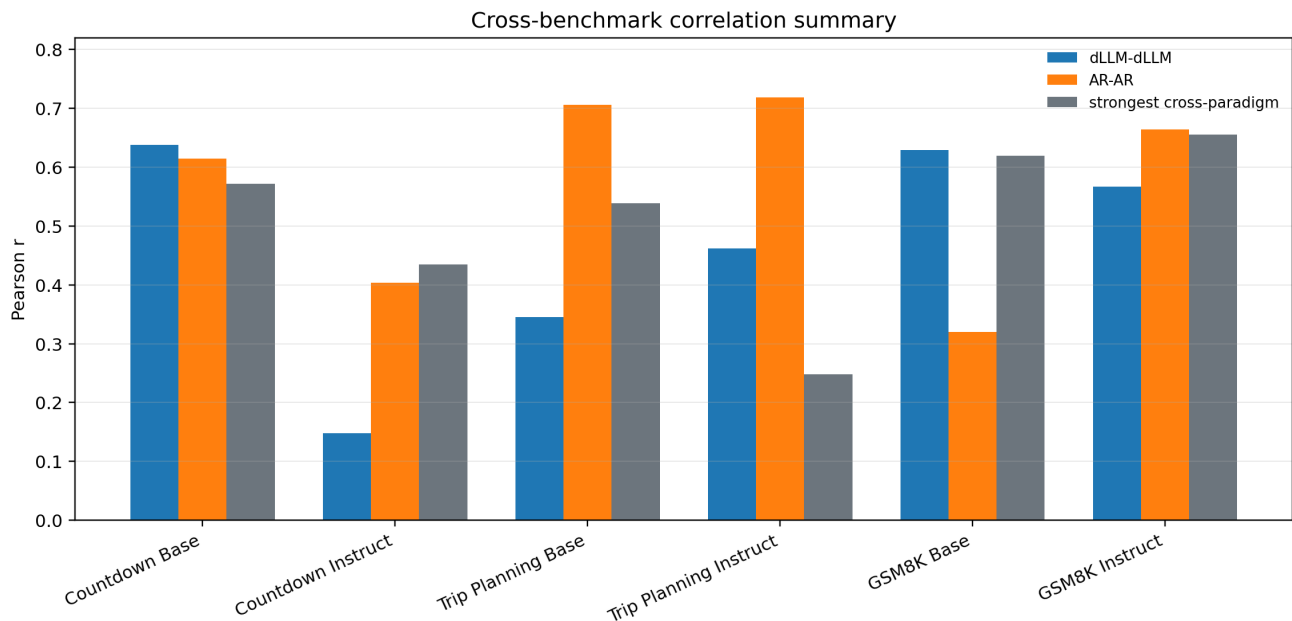
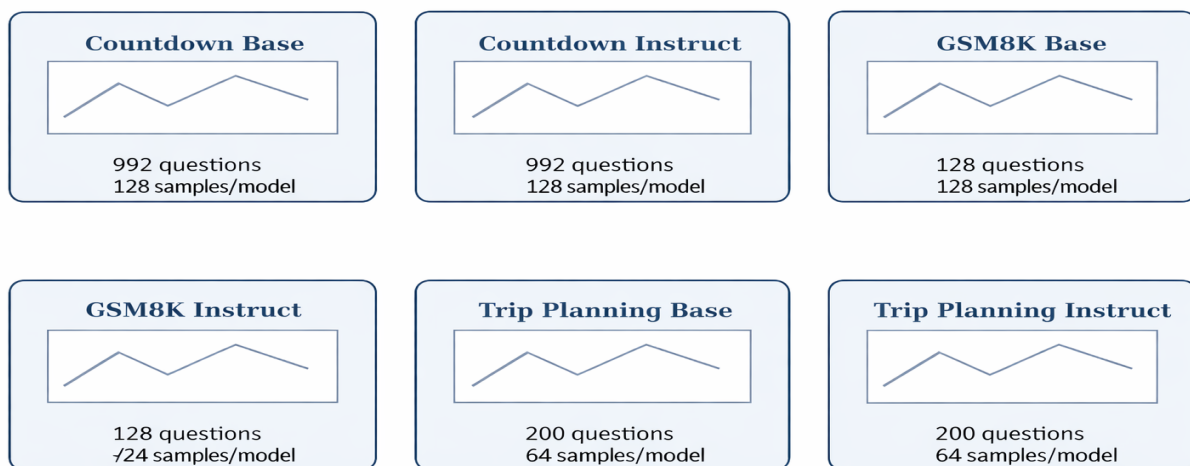


Figure 9. Cross-benchmark summary of within-paradigm and strongest cross-paradigm correlations. Diffusion models agree most on Countdown Base, autoregressive models agree most on Trip Planning, and GSM8K shows a more mixed pattern.

Placeholder layout for the six interactive HTML comparison views



Use exported screenshots from the HTML tool in the final thesis if per-question scatter plots are needed.

Figure 10. Placeholder layout for the six interactive HTML comparison views (R2). The milestone draft names six HTML files for per-question inspection; this placeholder records their role without fabricating scatter-plot data that are not included in the authoritative markdown draft.

Source: author placeholder based on MILESTONE_2_DRAFT.md Section 7.8.

Table 11. Countdown Base per-question correlation matrix.

Model Pair	Type	Pearson r	p-value
LLaDA-Base vs Dream-Base	dLLM-dLLM	0.6382	1.4×10^{-114}
Qwen-Base vs Llama-Base	AR-AR	0.6149	3.1×10^{-104}
Dream-Base vs Qwen-Base	dLLM-AR (shared arch)	0.5718	3.5×10^{-87}
LLaDA-Base vs Qwen-Base	dLLM-AR	0.5342	2.8×10^{-74}
LLaDA-Base vs Llama-Base	dLLM-AR	0.4580	1.3×10^{-52}
Dream-Base vs Llama-Base	dLLM-AR	0.4432	5.6×10^{-49}

Table 12. Countdown Instruct per-question correlation matrix.

Model Pair	Type	Pearson r	p-value
Dream-Inst vs Llama-Inst	dLLM-AR	0.4345	6.1×10^{-47}
Dream-Inst vs Qwen-Inst	dLLM-AR	0.4062	1.1×10^{-40}
Qwen-Inst vs Llama-Inst	AR-AR	0.4039	3.3×10^{-40}
LLaDA-Inst vs Dream-Inst	dLLM-dLLM	0.1475	3.1×10^{-6}
LLaDA-Inst vs Llama-Inst	dLLM-AR	0.1464	3.6×10^{-6}
LLaDA-Inst vs Qwen-Inst	dLLM-AR	0.0867	6.3×10^{-3}

Table 13. GSM8K Base per-question correlation matrix.

Model Pair	Type	Pearson r	p-value
LLaDA-Base vs Dream-Base	dLLM-dLLM	0.6291	1.8×10^{-15}
Dream-Base vs Qwen-Base	dLLM-AR (shared arch)	0.6189	6.9×10^{-15}
LLaDA-Base vs Qwen-Base	dLLM-AR	0.4907	4.1×10^{-9}
LLaDA-Base vs Llama-Base	dLLM-AR	0.4316	3.7×10^{-7}
Dream-Base vs Llama-Base	dLLM-AR	0.3510	4.9×10^{-5}
Qwen-Base vs Llama-Base	AR-AR	0.3204	2.3×10^{-4}

Table 14. GSM8K Instruct per-question correlation matrix.

Model Pair	Type	Pearson r	p-value
Qwen-Inst vs Llama-Inst	AR-AR	0.6638	1.4×10^{-17}
LLaDA-Inst vs Llama-Inst	dLLM-AR	0.6550	5.0×10^{-17}
Dream-Inst vs Llama-Inst	dLLM-AR	0.6398	4.3×10^{-16}
LLaDA-Inst vs Dream-Inst	dLLM-dLLM	0.5671	3.0×10^{-12}
Dream-Inst vs Qwen-Inst	dLLM-AR (shared arch)	0.5598	6.5×10^{-12}

Model Pair	Type	Pearson r	p-value
LLaDA-Inst vs Qwen-Inst	dLLM-AR	0.4847	6.8×10^{-9}

Table 15. Trip Planning Base per-question correlation matrix.

Model Pair	Type	Pearson r	p-value
Qwen-Base vs Llama-Base	AR-AR	0.7061	1.7×10^{-31}
LLaDA-Base vs Llama-Base	dLLM-AR	0.5394	1.7×10^{-16}
LLaDA-Base vs Qwen-Base	dLLM-AR	0.5158	5.5×10^{-15}
Dream-Base vs Qwen-Base	dLLM-AR (shared arch)	0.4829	4.4×10^{-13}
Dream-Base vs Llama-Base	dLLM-AR	0.3629	1.3×10^{-7}
LLaDA-Base vs Dream-Base	dLLM-dLLM	0.3451	5.6×10^{-7}

Table 16. Trip Planning Instruct per-question correlation matrix.

Model Pair	Type	Pearson r	p-value
Qwen-Inst vs Llama-Inst	AR-AR	0.7190	4.1×10^{-33}
LLaDA-Inst vs Dream-Inst	dLLM-dLLM	0.4622	5.6×10^{-12}
Dream-Inst vs Llama-Inst	dLLM-AR	0.2484	3.9×10^{-4}
LLaDA-Inst vs Qwen-Inst	dLLM-AR	0.2170	2.0×10^{-3}
LLaDA-Inst vs Llama-Inst	dLLM-AR	0.1647	2.0×10^{-2}
Dream-Inst vs Qwen-Inst	dLLM-AR (shared arch)	0.1572	2.6×10^{-2}

Table 17. Summary of within-paradigm and strongest cross-paradigm correlations across benchmark-variant conditions.

Benchmark	dLLM-dLLM (r)	AR-AR (r)	Strongest cross-paradigm (r)
Countdown Base	0.638	0.615	Dream-Qwen: 0.572
Countdown Instruct	0.148*	0.404	Dream-Llama: 0.435
Trip Planning Base	0.345	0.706	LLaDA-Llama: 0.539
Trip Planning Instruct	0.462	0.719	Dream-Llama: 0.248
GSM8K Base	0.629	0.320	Dream-Qwen: 0.619
GSM8K Instruct	0.567	0.664	LLaDA-Llama: 0.655

- Finding 1: On Countdown Base, the strongest correlation is between the two diffusion models: LLaDA-Base versus Dream-Base reaches $r = 0.6382$, which is higher than Dream-Base versus Qwen-Base at $r = 0.5718$ even though Dream shares the Qwen2.5 backbone. This is the strongest evidence that generation paradigm matters more than pretrained weights for failure-mode alignment in this setting.

- Finding 2: On Trip Planning, the AR-AR pair is strongest in both Base and Instruct settings. Qwen-Base versus Llama-Base reaches $r = 0.7061$, and Qwen-Inst versus Llama-Inst reaches $r = 0.7190$.
- Finding 3: Trip Planning Instruct shows extremely low cross-paradigm agreement. Dream-Inst versus Qwen-Inst reaches $r = 0.1572$ and LLaDA-Inst versus Llama-Inst reaches $r = 0.1647$, which means the two paradigms solve almost disjoint subsets of problems.
- Finding 4: GSM8K is mixed and partly saturated. On GSM8K Base, the diffusion pair is strongest at $r = 0.6291$, but on GSM8K Instruct the AR pair leads at $r = 0.6638$ and several cross-paradigm correlations are also high.
- Finding 5: LLaDA-Instruct is an outlier on Countdown. Its Countdown Instruct correlations range from 0.0867 to 0.1475, consistent with its collapsed pass@128 performance of 5.54%.
- Finding 6: The benchmark dependence is the synthesis result. Diffusion models agree most on Countdown, autoregressive models agree most on Trip Planning, and GSM8K shows the weakest paradigm separation.

8. Provisional take-home messages

- Evaluation protocol determines the headline ranking. Diffusion models are best on deterministic or pass@1 planning, but autoregressive models win once k becomes large enough.
- Per-question agreement patterns are a genuine scientific result, not just a visualization convenience. On Countdown Base, paradigm explains failure-mode alignment better than shared pretrained architecture; on Trip Planning Instruct, the low cross-paradigm correlations point to strong ensemble potential.
- Prompting is part of the scientific question. Diffusion models are much stronger at zero-shot GSM8K, while AR models gain more from few-shot prompting and maintain broader sample diversity under pass@ k .

9. Scope and limitations

This milestone is strongest when framed as a outline with complete evidence, not as the final thesis chapter. The main story is now coherent and numerically complete.

- All main dLLM comparisons use fast-dLLM inference, while all main AR comparisons use vLLM. Standard dLLM and AR-HF runs are kept only for targeted deterministic validation.
- Llama-3.1-8B is now included as the fourth model throughout the milestone document.
- Correlation analysis uses 128 samples/model on Countdown and GSM8K and 64 samples/model on Trip Planning.
- Trip Planning generation files for Dream, Qwen, and LLaDA come from an earlier run batch; Llama was added later in the Milestone 2 refresh.

10. References

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